Emilio Garcia Rabell

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Education

Bachelor of Science, Computer Engineering (Minor in Mathematics) August 2022 - May 2027
Rochester Institute of Technology GPA: 3.3

Skills

Embedded Systems: C++, C, Python, ESP32, Raspberry Pi, I2C, SPI, UART.
Systems & Hardware: Linux, sensors and actuators, hardware debugging, test automation, data acquisition.
Hardware Design & Validation: Computer architecture, FPGA, VHDL, Verilog, oscilloscopes, multimeters.
Tools: Git, Docker, CI/CD, cross-compilation, Vivado, ModelSim.

Experience

RIT - DIRS LAB Rochester, NY - *Research Assistant & Embedded Software Engineer* *May 2026 - Present*
- Develop and debug multithreaded C++ software for a robotic measurement platform integrating two Raspberry Pis, an x86 SBC, motorized axes, and spectrometers over a distributed Linux system.
- Diagnose hardware/software communication and system-level failures and build Docker-based cross-compilation and CI/CD pipelines for repeatable embedded Linux builds, deployment, and testing.

WHOOP Boston, MA - *Manufacturing Test Engineer Co-op* *July 2025 - December 2025*
- Developed Python validation and diagnostic tools that communicated with wearable device hardware over BLE and analyzed firmware telemetry to isolate hardware and firmware failures.
- Built a diagnostic application combining manufacturing test results and device telemetry to identify likely hardware root causes, reducing failure analysis time by 60%.

SharkNinja Needham, MA - *Product Test Engineer Co-op* *June 2024 - December 2024*
- Developed C++ firmware and software for Arduino and Raspberry Pi-based test systems, integrating sensors and actuators to automate repeatable hardware validation.
- Built C++ and Python data-acquisition and analysis tools to characterize electromechanical device behavior and diagnose hardware failures during testing.

Novatec Group Leon, GTO - *Software Engineer Intern* *May 2023 - August 2023*
- Engineered a real-time Java data-ingestion system monitoring 30 industrial machines simultaneously and processing 5,000+ parameters per machine.

Projects

NXP Cup Autonomous Car - *Academic Project* *January 2026 - May 2026*
- Developed and debugged real-time C firmware on a TI MSPM0 microcontroller, processing line-scan sensor data into steering commands and tuning control behavior to achieve a 26.175s lap and 2nd-place finish.

Sleeper - *Personal Project* *May 2026 - Present*
- Built and debugged an ESP32-based embedded system in C/C++, integrating microphone acquisition, hardware interfaces, WebSocket communication, and server-side processing.

5-Stage Pipelined MIPS Processor - *Academic Project* *January 2025 - May 2025*
- Designed a 5-stage pipelined 32-bit MIPS processor in VHDL with datapath, control logic, forwarding, and hazard detection; developed ModelSim testbenches and deployed the design to a Basys3 FPGA using Vivado.

Leadership

Microsoft LEAD ASPIRE Challenge - *1st Place* *October 2026*
- Won 1st place on a 5-person team by developing and pitching Align Buddy, a Mastercard-focused solution for securing agentic purchases through customer preference and permission validation.

Bernard Kozel Startup Program at RIT - *Team Leader* *August 2026 - May 2027*
- Co-lead a team of 5 students developing OpenCTF, translating findings from 100+ customer discovery interviews into product and technical requirements.

SHPE at RIT Eboard Member - *Freshman Rep/Treasurer/Academic Director* *September 2022 - May 2025*
- Led academic and professional initiatives and raised over $15,000 annually to support national conference attendance for more than 10 members.